

Kaan Kale

Reliable, private, and efficient AI systems

PHD STUDENT AT GEORGIA TECH

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Profile

PhD student in Machine Learning at Georgia Tech working across verifiable reasoning, privacy-preserving representations, and efficient training. Research spans formalization for language-model reasoning, information-theoretic representation learning, embedding reconstruction, and distributed AI systems. Experienced in large-scale PyTorch training and long-context model infrastructure.

Education

Georgia Tech

2026 – present

PhD in Machine Learning, School of Electrical and Computer Engineering

Atlanta, GA

Research: large language models, state space models, mechanistic interpretability, privacy, and efficient training.

Boğaziçi University

2020 – 2025

BSc in Electrical and Electronics Engineering; specialization in Signal Processing

Istanbul, Turkey

Research & Industry Experience

Protopia AI

May – Aug 2026

Research Engineer Intern

London, UK

- Built large-scale training and fine-tuning infrastructure for NVIDIA Nemotron-3 Ultra, implementing expert, context/sequence, tensor, and fully sharded data parallelism.
- Profiled long-context training memory usage and redesigned KNN metrics, logit materialization, and LM-head execution to support substantially longer training sequences.
- Investigated numerical instability in long-context Mamba/Transformer training and developed instrumentation for model states, training dynamics, and chunk-level utility metrics.
- Extended privacy-preserving LLM infrastructure through TurboQuant, output-protection and proxy integrations, VisionCloak evaluations, and shared-model training workflows.

Protopia AI

Nov 2024 – Dec 2025

Research Contractor

Remote

- Developed and evaluated learned transformations that protect sensitive information in language-model embeddings while preserving downstream utility.
- Developed BeamClean, a language-aware embedding reconstruction attack combining noise-parameter estimation with language-model-guided decoding.
- Contributed to the Stained Glass Transform through reconstruction attacks, mutual-information privacy metrics, utility analysis, and scalable PyTorch experiments.

University of Texas at Austin – LINC Lab

Aug 2023 – Dec 2024

Undergraduate Researcher; supervisor: Prof. Sriram Vishwanath

Austin, TX

- Researched privacy-aware and information-theoretic representation learning for language models and networked intelligent systems.

University of Texas at Austin – Swarm Lab

Jun 2023 – Dec 2024

Undergraduate Researcher; supervisor: Prof. Sandeep Chinchali

Austin, TX

- Developed learning and resource-allocation methods for heterogeneous fleets, speech systems, and privacy-aware distributed AI.

Technical Profile

Research

LLMs, privacy, mechanistic interpretability, information theory, representation learning, state space models

Programming

Python, C++, MATLAB, VHDL, LaTeX

ML systems

PyTorch, TensorFlow, distributed training, model parallelism, memory profiling

Honors & Beyond Research

Tekfen Holding Merit and Need-Based Scholarship (2020–present) · Ranked 301st of 2.2 million in Turkey’s national university entrance examination.

Piano performance studies at Istanbul University State Conservatory (2012–2021). DJing, music production, piano, kitesurfing, tennis, and golf. Languages: Turkish, English, and Italian.

Publications

Peer-reviewed papers and current preprints. Kaan Kale is shown in bold. ICLR 2026

ProofBridge: Auto-Formalization of Natural Language Proofs in Lean via Joint Embeddings

P. Jana, **K. Kale**, A. E. Tanriverdi, C. Song, S. Vishwanath, and V. Ganesh.

PREPRINT, 2026

RoboShape: Information-Theoretic Point Cloud Representations for Privacy-Aware Robot Perception

O. Baser, M. Sozen, **K. Kale**, S. Chinchali, and S. Vishwanath.

PREPRINT, 2026

VeryTrace: Verifying Reasoning Traces through Compilable Formalism and Structured Verification

N. Zhong, A. E. Tanriverdi, **K. Kale**, and S. Vishwanath.

ISIT 2026

MultiTok: Variable-Length Tokenization for Efficient LLMs Adapted from LZW Compression

N. Elias, H. Esfahanizadeh, **K. Kale**, S. Vishwanath, and M. Médard.

GLOBECOM 2025

Fair Resource Allocation for Fleet Intelligence

O. Baser, **K. Kale**, P.-H. Li, and S. Chinchali. IEEE Global Communications Conference, pp. 4952–4957.

PREPRINT, 2025

Learning Obfuscations of LLM Embedding Sequences: Stained Glass Transform

J. Roberts, K. Mylonakis, S. Roy, and **K. Kale**.

PREPRINT, 2025

BeamClean: Language Aware Embedding Reconstruction

K. Kale, K. Mylonakis, J. Roberts, and S. Roy.

INTERSPEECH 2025

WavShape: Information-Theoretic Speech Representation Learning for Fair and Privacy-Aware Audio Processing

O. Baser, A. E. Tanriverdi, **K. Kale**, S. Chinchali, and S. Vishwanath. Proceedings of Interspeech, pp. 4873–4877.

CoRL 2024

Fleet Supervisor Allocation: A Submodular Maximization Approach

O. Akcin, A. E. Tanriverdi, **K. Kale**, and S. Chinchali. 8th Annual Conference on Robot Learning.

ISIT 2024

TexShape: Information Theoretic Sentence Embedding for Language Models

K. Kale, H. Esfahanizadeh, N. Elias, O. Baser, M. Médard, and S. Vishwanath. IEEE International Symposium on Information Theory, pp. 2038–2043.

INTERSPEECH 2024

SecureSpectra: Safeguarding Digital Identity from Deep Fake Threats via Intelligent Signatures

O. Baser, **K. Kale**, and S. Chinchali. Proceedings of Interspeech, pp. 1115–1119.

Selected Research Projects

Passive Radar Using Ambient FM Signals

Sep 2024 – present

Aircraft detection using passive sensing and machine learning

- Developing a covert radar pipeline using ambient FM/TV signals, supervised by Prof. Ali Emre Pusane.

LLM Privacy / Coded LLM

Aug 2023 – Aug 2024

Privacy-preserving and efficient language-model pipelines

- Developed information-theoretic methods for private, efficient language-model representation under Prof. Sriram Vishwanath and Prof. Muriel Médard.